APPLIED MULTIVARIATE METHODS
833:525 / 970:527
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Problem 1:

Dataset to be used: `workdata.csv`

Administrators study the attendance behavior of workers at two business locations for one year (hint: the offset is the same for all workers and is equal to 260 days). Predictors of absence rate include job type and standardized test score in a professional exam. The data are defined:

Variable	Name	Code
Location	loc	1=Location A, 0=Location B
Job type	type	1=administrative, 0=non-administrative
Exam score	score	Standardized test score in professional exam
Absence	daysabs	Number of days absent in calendar year
Days worked	dayswk	Number of days scheduled to work in calendar year

a) Fit a Poisson regression model to study the association between job type and work absence rate. Write the population model. (10 points)

Population Model: log (Expected absent days/260) = β0 + β1 * Xtype

Expected absent days: is the expected count of absent days for a worker
Working days: is the offset for this Poisson regression, it is the total number of days of scheduledwork in a year, in this case it is 260 days.

- X_{type} is the independent binary predictor for job type, it is 1 if the job type is administrative, and 0 for non-administrative job.
- β_0 (intercept) – it is the log of expected absence rate for a non-administrative employee. (when job type=0)
- β_1 (slope coefficient) – it is the log rate of absence per day worked comparing those whose job$_{type}$ is administrative versus those with a non-administrative job, holding all else constant. (when job type=1)

Fitted Model: log (Expected absent days/260) =–3.65894–-0.31729 * Xtype

- **β_0 (intercept)** = -3.65894, the expected log rate of absence per 260 working days for a person whose job type is non-administrative is -3.65894.
- **(Expected Rate of absence for non-Admin job)**: $e^{-3.65894}$ is the expected rate of absence per workyear for a non-administrative staff member.
- **β_1 (slope coefficient)** = -0.31729, the change in log rate of absence per 260 working days for a person whose job type is administrative decreases by 0.31729.
- **(Expected Rate of absence for Admin job)** The absence rate for administrativestaff decreases by $e^{-0.31729}$ compared to non-administrative staff.

```
R 4.3.2 · C:/Users/HP PAVILLION 17/OneDrive/Desktop/Multivariate/

Call:
glm(formula = daysabs ~ type, family = "poisson", data = data,
    offset = log(dayswk))

Coefficients:
            Estimate Std. Error  z value Pr(>|z|)
(Intercept) -3.65894    0.03036 -120.527  < 2e-16 ***
type1       -0.31729    0.04747   -6.684 2.32e-11 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

    Null deviance: 2409.8  on 315  degrees of freedom
Residual deviance: 2364.5  on 314  degrees of freedom
AIC: 3229.9
```

b) Give an estimate and 95% CI of the absence rate ratio between administrative and non-administrative workers. Interpret the estimate and the 95% CI. Make a conclusion about statistical significance and provide evidence for your conclusion. (10 points)

In mathematical sense these are the exponential values of log rates and log rate ratios which we did for part a it is also doing the statistical test through confidence interval

```
> rate_ratio_model <- exp(coef(model))
> conf_int1 <- exp(confint(model))
Waiting for profiling to be done...
> cbind(RR = rate_ratio_model, conf_int1)
                    RR     2.5 %     97.5 %
(Intercept) 0.02575973 0.0242572 0.02732302
type1       0.72812257 0.6632512 0.79891872
>
```

- Advantage of Rate ratio (RR): Rate ratio makes better sense for the sake of interpretation of results for researchers and readers
- The Rate Ratio (RR) for type1 (administrative workers compared to non-administrativeworkers) is **0.72812257.** This means that the expected absence rate for administrative workers is about **72.8%** of the rate for non-administrative workers, indicating that administrative workers are expected to have fewer absences.
- Alternatively, we can also say that administrative workershave a **27.2%** lower rate of absence compared to non-administrative workers, after adjusting for the number of days worked. This is a significant finding that suggests job type is an important factor in predicting absence rates.
- The 95% CI for type1 ranges from **0.6632512 to 0.79891872**. This interval suggests thatwe are 95% confident that the true absence rate ratio (administrative to non- administrative workers) lies within this range. One can also see that Rate Ratio does not include 1 at 95% CI, means results of rate ratios are statistically significance for our analysis.

c) Fit two Poisson regression models: (M1) using location and job type as predictors and (M2) using standardized test score, location, and job type as predictors. Is M2 significantly better than M1? Write down the null and alternative hypotheses, test statistic, p value, decision, and conclusion. (10 points)

- **Null Hypothesis (H_o):** Reduced model is better (M1) i.e., the standardized test score (score) as an additional predictor does not fit the model, ***H*0: *β*score=0,**
- **Alternative Hypothesis (H_1):** Full model is better i.e. Model M2, which includes the standardized test score as a predictor, provides a significantly better fit to the data than Model M1. ***β*score≠0**
- **Test statistic:** used to compare the models is the change in residual deviance, which follows a chi-square distribution: ΔDeviance=7.3221 with ΔDf=1. The p-value obtained from the chi-square test for this difference in deviance.
- p=0.006811. Given that the p-value is less than the conventional alpha level (α=0.05), we reject the null hypothesis in favor of the alternative. It means Full mode is better i.e. M2 is better than M1. This finding is statistically significant, as evidenced by the p-value, and supports the inclusion of the standardized test score in the model.

```
> # annvova test comparisons for two models will give us whether tests score is significant predictor to include (longer mode is better or reduce mod
el is better)
> anova_two_models <- anova(M1, M2, test = "Chisq")
> # Results of comparisons
> anova_comparison
Analysis of Deviance Table

Model 1: daysabs ~ type + loc
Model 2: daysabs ~ type + loc + score
  Resid. Df Resid. Dev Df Deviance Pr(>Chi)
1       313     2029.3
2       312     2022.0  1   7.3221 0.006811 **
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

d) Provide and interpret the estimate and 95% CI for standardized test score in professional exam after adjustment for job type and location. Make a conclusion about statistical significance and provide evidence for your conclusion. (10 points)

```
> rate_ratio_model2 <- exp(coef(M2))
> conf_int2 <- exp(confint(M2))
Waiting for profiling to be done...
> cbind(RR = rate_ratio_model2, conf_int2)
                    RR      2.5 %     97.5 %
(Intercept) 0.01907163 0.01604653 0.02263521
type1       0.66457130 0.60511184 0.72949481
loc1        2.33551141 2.09934276 2.60112548
score       0.99631784 0.99365193 0.99898476
>
```

- **Rate Ratio for test scores:**

The rate ratio for standardized test scores is 0.9963, which come after the exponential of log rate ratio of test scores. This indicates that for each point increase in the standardized test score, the absence rate is expected to decrease by only 0.37%, adjusting for jobtype and location, holding all else constant.

- **Confidence Interval:**

At a 5% level of significance, the confidence interval for the rate ratio of the standardized testscore is 0.993 to 0.998, this doesn't include 1, it means at 95 percent confidence results show that the association is statistically significant.

R-Code:

```
data <- read.csv("/Users/imanbasit/Downloads/workdata.csv")
data$type <- factor(data$type)
data$loc <- factor(data$loc)
data$type <- relevel(data$type, ref = "0")
data$loc <- relevel(data$loc, ref = "0")
# Fit a Poisson regression model
model <- glm(daysabs ~ type, family = "poisson", data = data, offset = log(dayswk))
# Display the summary of the model
summary(model)
rate_ratio_model <- exp(coef(model))
conf_int1 <- exp(confint(model))
cbind(RR = rate_ratio_model, conf_int1)
# Model M1 includes location and job type as predictors
M1 <- glm(daysabs ~ type + loc, family = "poisson", data = data, offset = log(dayswk))
summary(M1)
#However, Model M2 has standardized test score, location, and job type as predictors, one additional predictor
M2 <- glm(daysabs ~ type + loc + score, family = "poisson", data = data, offset = log(dayswk))
summary(M2)
# annvova test comparisons for two models will give us whether tests score is significant predictor to include (longer mode is
better or reduce model is better)
anova_two_models <- anova(M1, M2, test = "Chisq")
# Results of comparisons
anova_comparison
rate_ratio_model2 <- exp(coef(M2))
conf_int2 <- exp(confint(M2))
cbind(RR = rate_ratio_model2, conf_int2)
```